Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 1: Large Numbers Around Us

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	1 — Large Numbers Around Us
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Every wrong option is the number you land on after a genuine slip — swapping one lakh for one million, shifting a comma by one period, confusing place value with face value, rounding the wrong way at a 5, or miscounting digits by one; at this level each question chains two or more such steps, so re-check the period of every comma and the direction of every rounding before you commit. Do not write on this paper — mark answers on your answer sheet.

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1. A 9-digit number has 6 in the crores place, 6 in the ten-thousands place, and 6 in the ones place; every other place is 0. The number is:

(A) 6,06,00,006 (B) 60,06,00,006
(C) 6,00,60,006 (D) 6,00,06,006

2. Using each of the digits 0, 3, 6, 8, 9 exactly once, let G be the largest possible 5-digit number and S the smallest 5-digit number (no leading zero). The value of G − S is:

(A) 67,941 (B) 68,041
(C) 69,741 (D) 62,841

3. Rewrite the Indian number 7,00,80,005 in the International comma system.

(A) 7,00,80,005 (B) 70,080,005
(C) 700,800,05 (D) 70,08,0005

4. In the number 7,07,07,070, the place value of the leftmost 7 is how many times the place value of the rightmost 7?

(A) 10,00,000
(B) 1,00,000
(C) 1,00,00,000
(D) 1,000

5. How many ten-thousands are there in 4 crore 5 lakh?

(A) 405
(B) 40,500
(C) 450
(D) 4,050

6. The number name 'six crore six lakh sixty thousand six hundred' in figures is:

(A) 6,66,60,600
(B) 6,06,60,600
(C) 6,06,06,600
(D) 66,60,600

7. A scientist records a distance as 4 crore 8 lakh kilometres. An international journal must print it with International commas. It should read:

(A) 40,800,000
(B) 4,08,00,000
(C) 408,000,00
(D) 4,080,000

8. Using each of 1, 4, 5, 6, 0 exactly once, the smallest 5-digit number divisible by 5 (no leading zero) is:

(A) 10,465
(B) 14,560
(C) 10,456
(D) 14,650

9. In 7,80,05,060, the sum of the place value of the 8 and the place value of the 5 is:

(A) 80,50,000
(B) 85,00,000
(C) 13
(D) 80,05,000

10. Reading right to left, the Indian system places commas after 3 digits, then 2, then 2, ... In the 11-digit number 12345678901, the leftmost (final) comma falls so the number reads:

(A) 12,34,56,78,901
(B) 1,2345,67,89,01
(C) 1,23,45,67,89,01
(D) 123,456,789,01

11. Arrange in INCREASING order: 9.4 lakh, 0.92 million, 9,60,000, 89 ten-thousands.

(A) 0.92 million < 89 ten-thousands < 9.4 lakh < 9,60,000
(B) 89 ten-thousands < 0.92 million < 9.4 lakh < 9,60,000
(C) 89 ten-thousands < 9.4 lakh < 0.92 million < 9,60,000
(D) 9.4 lakh < 89 ten-thousands < 0.92 million < 9,60,000

12. One billion is how many crores?

(A) 1,000
(B) 10
(C) 100
(D) 10,000

13. Estimate 5,950 × 408 by rounding 5,950 to the nearest thousand and 408 to the nearest hundred, then state the estimate.

(A) 2,40,000
(B) 24,00,000
(C) 20,40,000
(D) 2,40,00,000

14. Round 6,49,500 to the nearest thousand, then round that result to the nearest ten-thousand, then to the nearest lakh. The three results in order are:

(A) 6,49,000 then 6,50,000 then 6,00,000(B) 6,50,000 then 6,50,000 then 6,00,000

(C) 6,49,000 then 6,40,000 then 6,00,000(D) 6,50,000 then 6,50,000 then 7,00,000

15. Without multiplying it out exactly, how many digits does the product 9,99,999 × 99 have?

(A) 8 (B) 7

(C) 8 or 9 (D) 9

16. A relief convoy must carry 1,450 sacks per truck for 7 trucks, and the depot should never run short. The most sensible round number of sacks to stock is:

(A) exactly 10,150 (B) round down to 10,000

(C) round down to 10,100 (D) round up to 11,000

17. The expanded form 4 × 1,00,00,000 + 6 × 1,00,000 + 9 × 1,000 + 3 × 10 represents:

(A) 40,60,09,030 (B) 4,06,09,030

(C) 4,06,90,030 (D) 4,60,09,030

18. A printing press runs 9,500 sheets an hour for a 24-hour day. Rounded to the nearest lakh, the day's output is:

(A) 2,00,000 (B) 1,00,000

(C) 3,00,000 (D) 2,30,000

19. Counting at one number every 3 seconds, about how long does it take to count to one lakh?

(A) a little over one day (B) about one week

(C) a little under 3 and a half days (D) about half a day

20. A district has 24 lakh people and the country it belongs to has 120 crore people. The district holds what fraction of the country's people?

(A) 1/50 (B) 1/5000

(C) 1/5 (D) 1/500

21. The number 80,80,808 read in words in the Indian system is:

(A) eight crore eight lakh eight hundred eight

(B) eighty lakh eighty thousand eight hundred eight

(C) eighty lakh eight thousand eight hundred eight

(D) eighty-eight lakh eight hundred eight

22. How many lakhs equal 8 arab?

(A) 8,000 (B) 8,00,000

(C) 80,000 (D) 800

23. In the International system, 600,070,008 read in words is:

(A) sixty crore seven lakh eight
(B) six hundred million seven thousand eight
(C) six hundred million seventy thousand eight
(D) six hundred sixty million seventy thousand eight

24. A digit has place value 50,00,000 in a number. A second digit with the SAME face value sits three places to its right. The ratio of the first digit's place value to the second's is:

(A) 100
(B) 1,000
(C) 10
(D) 10,000

25. A charity collected ₹3 crore 60 lakh and shared it equally among 24 villages. Each village received:

(A) ₹15,00,000
(B) ₹1,50,000
(C) ₹1,50,00,000
(D) ₹15,000

26. Which of these is the largest number?

(A) 81 million
(B) 8,15,00,000
(C) 8.2 crore
(D) 815 lakh

27. Round 7,49,52,000 to the nearest crore, then round the ORIGINAL number to the nearest ten-lakh. The two results are:

(A) 7,00,00,000 and 7,50,00,000
(B) 8,00,00,000 and 7,50,00,000
(C) 7,00,00,000 and 7,40,00,000
(D) 7,50,00,000 and 7,50,00,000

28. Estimate the sum 4,82,000 + 2,49,000 + 3,71,000 by rounding each addend to the nearest lakh first. The estimate is:

(A) 11,00,000
(B) 10,00,000
(C) 12,00,000
(D) 10,02,000

29. How many digits does the product 4,00,000 × 25,00,000 have?

(A) 12
(B) 11
(C) 13
(D) 14

30. A bus carries 52 passengers; a city runs 1,150 such buses, each making 8 trips a day. About how many passenger-trips happen daily? (Round 52 to 50 and 1,150 to 1,000 first.)

(A) about 40,000
(B) about 40,00,000
(C) about 4,00,000
(D) about 80,000

31. A reservoir held 2 crore litres. First 78,40,000 litres drained out, then 35,60,000 litres were pumped back in. The litres now in the reservoir are:

(A) 1,21,60,000
(B) 1,57,20,000
(C) 1,14,00,000
(D) 2,42,00,000

32. The number name 'eight crore eight thousand eight' in figures is:

(A) 8,08,008
(B) 8,00,80,008
(C) 80,08,008
(D) 8,00,08,008

33. The same number is 7,030,400 in the International system. Written with Indian commas it is:

(A) 7,03,0400
(B) 7,03,04,00
(C) 70,304,00
(D) 70,30,400

34. In 9,07,40,608, by how much does the place value of the leftmost digit exceed the place value of the digit 4?

(A) 8,60,00,000
(B) 8,96,00,000
(C) 5
(D) 8,99,60,000

35. Written with Indian commas, the 12-digit number 123456789012 becomes:

(A) 12,34,56,78,9012
(B) 123,456,789,012
(C) 1,23,45,67,89,012
(D) 1,2345,6789,012

36. A 5-digit palindrome (reads the same both ways) has its ten-thousands digit equal to 7 and its hundreds (middle) digit equal to 3. How many such palindromes exist?

(A) 10
(B) 9
(C) 100
(D) 1

37. Arrange in DECREASING order: 7.5 crore, 76 million, 7,55,00,000, 740 lakh.

(A) 76 million > 7,55,00,000 > 7.5 crore > 740 lakh
(B) 7,55,00,000 > 76 million > 7.5 crore > 740 lakh
(C) 740 lakh > 7.5 crore > 7,55,00,000 > 76 million
(D) 7.5 crore > 7,55,00,000 > 76 million > 740 lakh

38. Round 6,95,000 to the nearest ten-thousand, then round the ORIGINAL to the nearest lakh, then add the two rounded results. The total is:

(A) 13,90,000
(B) 14,00,000
(C) 13,00,000
(D) 12,90,000

39. Which is the correct expanded form of 5,09,070?

(A) 5 × 1,00,000 + 0 × 10,000 + 9 × 1,000 + 0 × 100 + 7 × 10 + 0 × 1
(B) 5 × 1,00,000 + 9 × 10,000 + 7 × 100
(C) 5 × 10,00,000 + 9 × 1,000 + 7 × 10
(D) 5 × 1,00,000 + 9 × 1,000 + 7 × 100

40. How many digits does the product of the largest 3-digit number and the smallest 5-digit number have?

(A) 8
(B) 9
(C) 7
(D) 5

41. How many digits does the product 8,00,000 × 8,00,000 have?

(A) 11
(B) 13
(C) 12
(D) 10

42. How many digits does the product 50,000 × 19,999 have?

(A) 10
(B) 8
(C) 11
(D) 9

43. How many digits does the product of the smallest 6-digit number and the smallest 4-digit number have?

(A) 9
(B) 8
(C) 10
(D) 24

44. Compare 8 lakh × 12 with 1 crore. Which statement is true?

(A) they are equal
(B) 8 lakh × 12 is larger (it is 1 crore 6 lakh)
(C) 1 crore is exactly ten times larger
(D) 1 crore is larger (8 lakh × 12 is 96 lakh)

45. Two cities have populations 0.65 crore and 72 lakh. How much larger is the second city's population?

(A) 7 thousand
(B) 7 lakh
(C) 70 lakh
(D) 60 lakh

46. A factory's output rose from 6,40,000 units to 1 crore 12 lakh units. The increase, rounded to the nearest ten-lakh, is:

(A) 1,00,00,000
(B) 1,10,00,000
(C) 90,00,000
(D) 1,06,00,000

47. A library has 7,60,000 books and triples its collection, then rounds the new total to the nearest ten-lakh for a banner. The banner figure is:

(A) 22,80,000
(B) 23,00,000
(C) 30,00,000
(D) 20,00,000

48. Three readings are 7,00,070, 7,07,000 and 70,007. By how much does the largest exceed the smallest?

(A) 6,30,063
(B) 6,36,993
(C) 6,99,930
(D) 7,36,993

49. A water plant pumps 2,45,000 litres per hour. About how many litres in one full day? (Round the hourly rate to the nearest lakh first.)

(A) about 4,80,000
(B) about 4,80,00,000
(C) about 60,00,000
(D) about 48,00,000

50. The smallest 11-digit number and the largest 10-digit number differ by:

(A) 10
(B) 1,00,00,00,000
(C) 1
(D) 0

51. How many lakhs are there in 60 million?

(A) 60
(B) 6,000
(C) 6
(D) 600

52. Three numbers are 5,06,000, 5,00,600 and 5,60,000. Their correct INCREASING order is:

(A) 5,00,600 < 5,60,000 < 5,06,000 (B) 5,00,600 < 5,06,000 < 5,60,000

(C) 5,06,000 < 5,00,600 < 5,60,000 (D) 5,60,000 < 5,06,000 < 5,00,600

53. A printing error wrote 5,40,000 as 54,000. By what factor is the wrong number too small, and what is the absolute error?

(A) 100 times too small; error 4,86,000 (B) 10 times too small; error 4,86,00,000

(C) 10 times too small; error 5,40,000 (D) 10 times too small; error 4,86,000

54. Mumbai's population is about 1 crore 28 lakh and Bengaluru's is about 84 lakh. About how much larger is Mumbai's population, rounded to the nearest ten-lakh?

(A) 40,00,000 (B) 44,00,000

(C) 4,00,000 (D) 50,00,000

55. A donor pledged 1 crore and gives it in equal weekly instalments over exactly 50 weeks. After 18 weeks, the amount still owed is:

(A) ₹36,00,000 (B) ₹64,00,000

(C) ₹82,00,000 (D) ₹6,40,000

56. Estimate 4,950 × 6,050 by rounding each factor to the nearest thousand, then compare with the exact product. The estimate and the comparison are:

(A) estimate 3,00,00,000; the estimate is less than the exact product

(B) estimate 30,00,000; the estimate is more than the exact product

(C) estimate 3,00,00,000; the estimate equals the exact product

(D) estimate 3,00,00,000; the estimate is more than the exact product

57. A stadium holds 65,000 spectators and is filled for 16 matches in a season. Rounded to the nearest lakh, the total attendance is:

(A) 10,00,000 (B) 11,00,000

(C) 1,00,000 (D) 10,40,000

58. A report states 3,49,500 visitors and rounds it to the nearest lakh; another states 3,50,001 visitors and rounds the same way. The two rounded figures are:

(A) 3,00,000 and 3,00,000 (B) 3,00,000 and 4,00,000

(C) 4,00,000 and 4,00,000 (D) 4,00,000 and 3,00,000

59. In 6,03,50,070, the place value of the 3 is divided by the face value of the 5. The result is:

(A) 6,00,000 (B) 60,00,000

(C) 60,000 (D) 12,000

60. A town of about 5 lakh people has roughly one doctor for every 1,250 people. Using these round figures, about how many doctors serve the town?

(A) about 4,000

(B) about 40

(C) about 400

(D) about 625

Solutions — Mathematics (Class 7), Chapter 1: Large Numbers Around Us — Paper 5

Paper 5 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	B	31	B	41	C	51	D
2	A	12	C	22	C	32	D	42	D	52	B
3	B	13	B	23	C	33	D	43	A	53	D
4	A	14	D	24	B	34	D	44	D	54	A
5	D	15	A	25	A	35	C	45	B	55	B
6	B	16	D	26	C	36	A	46	B	56	D
7	A	17	B	27	A	37	A	47	D	57	A
8	A	18	A	28	A	38	B	48	B	58	B
9	D	19	C	29	C	39	A	49	D	59	C
10	C	20	D	30	C	40	C	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Crore place value is 1,00,00,000, ten-thousand is 10,000, ones is 1. So the number is 6 x 1,00,00,000 + 6 x 10,000 + 6 x 1 = 6,00,00,000 + 60,000 + 6 = 6,00,60,006, a 9-digit number. Placing the second 6 in the lakhs place instead of ten-thousands gives 6,06,00,006. Reading the first 6 as ten-crores makes a 10-digit number 60,06,00,006. Putting the 60,000 as 6,000 (thousands place) collapses it to 6,00,06,006.

2. (A) Largest: place the digits high-to-low, G = 98,630. Smallest: the smallest non-zero digit must lead (3), then the rest including 0 in ascending order, S = 30,689. Difference = 98,630 – 30,689 = 67,941. Letting a 0 lead the smallest (writing 03,689) breaks the no-leading-zero rule and inflates the difference toward 68,041. Mis-ordering the smallest as 30,869 gives 67,761-type slips, while putting 9 before 8 in G (98,360) shifts the result toward 69,741. Reversing two digits of the smallest to 36,089 gives 62,841.

3. (B) Strip commas: 70080005, which is 8 digits. Group in threes from the right: 70,080,005. Keeping the Indian commas is no conversion at all. Grouping the leftover digits on the right as 700,800,05 violates the 3-3-3 rule. The split 70,08,0005 uses a 2-2-4 pattern belonging to neither system.

4. (A) The four 7s sit in the crores, lakhs, thousands and tens places. The leftmost 7 is in the crores place: place value 7,00,00,000. The rightmost 7 is in the tens place: place value 70. The ratio is 7,00,00,000 ÷ 70 = 10,00,000 (ten lakh). Counting one position too few (treating the gap as five tens-steps) gives 1,00,000. Counting one too many gives 1,00,00,000. Comparing only against the thousands 7 instead of the tens 7 gives 1,000.

5. (D) 4 crore 5 lakh = 4,05,00,000. One ten-thousand = 10,000, so the count is 4,05,00,000 ÷ 10,000 = 4,050. Dividing by one lakh (1,00,000) instead gives 405. Dividing by one thousand gives 40,500. Reading 4 crore 5 lakh as 4,50,00,000 then ÷ 10,000 gives 4,500-type errors; treating it as 45 lakh gives 450.

6. (B) Crores 6, then 'six lakh sixty thousand six hundred' = 6,60,600, so the full number is 6,06,60,600. Reading 'six lakh' into the ten-lakh slot inflates it to 6,66,60,600. Writing 'sixty thousand' as 06 thousand gives 6,06,06,600. Dropping the empty lakhs/ten-lakhs structure collapses it to the 8-digit 66,60,600.

7. (A) 4 crore 8 lakh = 4,08,00,000 (Indian). Strip commas: 40800000, 8 digits. Group in threes from the right: 40,800,000. Keeping Indian commas (4,08,00,000) is no conversion. The grouping 408,000,00 breaks the 3-3-3 rule. Reading the count as only 40 lakh 80 thousand gives 4,080,000, ten times too small.

8. (A) Divisible by 5 means the number must end in 0 or 5. To be smallest, the leading digit is the smallest non-zero digit, 1, and we want small digits next. Ending in 0 forces the 0 into the units place, leaving 4,5,6 for the middle in ascending order → 14,560. Ending in 5 lets the 0 sit just after the lead (allowed since it is not the leading digit): 1,0,then 4,6 ascending, then 5 → 10,465. Since 10,465 < 14,560, the smallest is 10,465. Forgetting that 0 may follow the lead gives 14,560. The number 10,456 does not end in 0 or 5, so it is not divisible by 5. 14,650 wastes the small 0 in the middle and is larger.

9. (D) Digits: crores 7, ten-lakhs 8, lakhs 0, ten-thousands 0, thousands 5, hundreds 0, tens 6, ones 0. The 8 is in the ten-lakhs place: place value 80,00,000. The 5 is in the thousands place: place value 5,000. Their sum is 80,00,000 + 5,000 = 80,05,000. Reading the 5 as ten-thousands gives 80,50,000. Reading the 8 as ten-lakhs but the 5 as lakhs gives 85,00,000. Adding face values 8 + 5 gives 13.

10. (C) Group from the right: first 3 → 901, then 2 → 89, then 2 → 67, then 2 → 45, then 2 → 23, then 2 → 1. So 1,23,45,67,89,01. Using 2-then-3 from the right (international style applied to Indian) gives 12,34,56,78,901. Mixing a 4-digit lakh group breaks it to 1,2345,... . Pure 3-3-3 grouping (International) gives 123,456,789,01.

11. (B) Convert all to plain numbers: 9.4 lakh = 9,40,000; 0.92 million = 0.92 × 10,00,000 = 9,20,000; 9,60,000 = 9,60,000; 89 ten-thousands = 89 × 10,000 =

8,90,000. Increasing: 8,90,000 < 9,20,000 < 9,40,000 < 9,60,000, i.e. 89 ten-thousands < 0.92 million < 9.4 lakh < 9,60,000. Treating 0.92 million as 92,000 mis-sorts it below the rest. Reading 89 ten-thousands as 8,90,000 is correct but swapping it past 0.92 million is the slip. Reading 9.4 lakh as 9,04,000 mis-orders it to the bottom.

12. (C) One billion = 1,00,00,00,000 (1 followed by 9 zeros) = 100 crore, since one crore = 1,00,00,000 (7 zeros) and 1,00,00,00,000 ÷ 1,00,00,000 = 100. Treating a billion as 10 followed by extra zeros (mistaking it for arab-scale) gives 1,000. Confusing billion with 100 crore's tenth gives 10. Reading billion as a lakh-crore gives 10,000.

13. (B) Round 5,950 to the nearest thousand → 6,000. Round 408 to the nearest hundred → 400. Product 6,000 × 400 = 24,00,000. Rounding 5,950 down to 5,000 and 408 to 400 gives 20,00,000, but careless digit-counting yields 2,40,000 (one zero short). Keeping 408 as ~408 and multiplying 6,000 × 408 ≈ 24,48,000 then misplacing a comma gives 20,40,000. Adding an extra zero gives 2,40,00,000.

14. (D) 6,49,500 to nearest thousand: the hundreds digit is 5, round up → 6,50,000. That to nearest ten-thousand: the thousands digit is 0 and the figure is already 6,50,000; the ten-thousands digit is 5 and thousands are 0, so it stays 6,50,000 (already a multiple of ten-thousand). That to nearest lakh: the ten-thousands digit is 5, round up → 7,00,000. Rounding the original DOWN at the 5 gives 6,49,000 chains. Treating 6,50,000 as rounding down to 6,00,000 at the lakh stage ignores that 50,000 rounds up.

15. (A) 9,99,999 is just under 10,00,000 and 99 is just under 100, so the product is just under 10,00,000 × 100 = 10,00,00,000 (an 8-digit number) and clearly more than 1,00,000 × 10 = 10,00,000. Exactly, 999999 × 99 = 999999 × 100 − 999999 = 99999900 − 999999 = 98999901, an 8-digit number. Assuming a 6-digit times a 2-digit always gives 6 + 2 − 1 = 7 digits is the slip giving 7. The product is unambiguously 8 digits, not a 7-or-9 range and not 9.

16. (D) Need = 1,450 × 7 = 10,150 sacks. To be SURE of never running short you must stock at least the exact need, so any sensible round figure must be ≥ 10,150 — rounding UP to 11,000 guarantees enough. Stocking exactly 10,150 leaves no margin and isn't a round figure for planning. Rounding down to 10,000 or 10,100 leaves you 150 (or 50) short, which fails the 'never run short' requirement.

17. (B) 1,00,00,000 is the crore unit → 4 crore = 4,00,00,000. 6 × 1,00,000 = 6,00,000 (6 lakh). 9 × 1,000 = 9,000. 3 × 10 = 30. Sum: 4,00,00,000 + 6,00,000 + 9,000 + 30 = 4,06,09,030. Treating 1,00,00,000 as ten-crore inflates the lead to 40,60,09,030. Putting the 9,000 into the ten-thousands group gives 4,06,90,030. Reading 6 × 1,00,000 as ten-lakh gives 4,60,09,030.

18. (A) Output = 9,500 × 24 = 2,28,000 sheets. To the nearest lakh: the lakhs digit is 2 and the part below the lakh is 28,000, which is less than 50,000, so round down to

2,00,000. Mis-multiplying to around 1,40,000 sheets and rounding gives 1,00,000. Rounding the 28,000 up wrongly gives 3,00,000. Rounding only to the nearest ten-thousand gives 2,30,000.

19. (C) One lakh = 1,00,000 numbers, each taking 3 seconds → 3,00,000 seconds. One day = 24 × 60 × 60 = 86,400 seconds. 3,00,000 ÷ 86,400 ≈ 3.47 days, which is a little under three and a half days. Ignoring the factor of 3 (using 1 second each) gives a little over one day. Overestimating the total to around six lakh seconds gives about a week. Using a wrongly large value for a day gives about half a day.

20. (D) Country = 120 crore = 1,20,00,00,000. District = 24 lakh = 24,00,000. Fraction = 24,00,000 ÷ 1,20,00,00,000 = 24 ÷ 12,000 = 1/500. Dividing 24 lakh by 12 crore-as-1,20,00,000 (a comma slip dropping a period) gives 1/50. Treating the country as 1,200 crore gives 1/5000. Mixing 24 lakh with 120 lakh gives 1/5.

21. (B) Group it: 80,80,808 → ten-lakhs 8, lakhs 0, ten-thousands 8, thousands 0, hundreds 8, tens 0, ones 8. That is 80 lakh, 80 thousand, 8 hundred 8 = eighty lakh eighty thousand eight hundred eight. Reading the leading 80 as a crore-scale group gives 'eight crore...'. Misreading the 80 thousand as 8 thousand drops a ten. Merging the lakh groups as 88 lakh ignores the empty thousands structure.

22. (C) 1 arab = 1,00,00,00,000 (1 followed by 9 zeros) = 1,000 crore. One lakh = 1,00,000, so 1 arab = 1,00,00,00,000 ÷ 1,00,000 = 10,000 lakh. Then 8 arab = 8 × 10,000 = 80,000 lakh. Treating 1 arab as only 100 crore gives 8,000. Over-counting a period gives 8,00,000. Treating 1 arab as 100 lakh-scale gives 800.

23. (C) Group: 600 | 070 | 008. The millions group is 600 (six hundred million), the thousands group is 070 (seventy thousand), the ones group is 008 (eight). So 'six hundred million seventy thousand eight'. The Indian reading 'sixty crore seven lakh eight' is the wrong system. Reading 070 as 7 thousand drops a ten. Misreading 600 as 660 inflates the millions.

24. (B) Each place to the right divides the unit weight by 10. Three places to the right means the weight is divided by 10 × 10 × 10 = 1,000. So the ratio of place values is 1,000 to 1. Counting only two places gives 100. Counting one place gives 10. Over-counting to four places gives 10,000.

25. (A) Total = 3 crore 60 lakh = 3,60,00,000. Divide by 24: 3,60,00,000 ÷ 24 = 15,00,000. So each village got ₹15,00,000 (fifteen lakh). Dividing 3,60,00,000 by 240 (an extra zero) gives ₹1,50,000. Mistaking the total as 36,00,00,000 and dividing gives ₹1,50,00,000. Dividing 3,60,00,000 by 2,400 gives ₹15,000.

26. (C) Convert all to plain numbers: 8.2 crore = 8,20,00,000; 81 million = 81 × 10,00,000 = 8,10,00,000; 8,15,00,000 = 8,15,00,000; 815 lakh = 815 × 1,00,000 = 8,15,00,000. The largest is 8,20,00,000 (8.2 crore). Treating 81 million as 8,10,00,000 is correct but it is smaller. Note 8,15,00,000 and 815 lakh are equal — both smaller than 8.2 crore. Reading 81 million as 8.1 crore (correct) still loses to 8.2 crore.

27. (A) Nearest crore: the lakhs-and-below part is 49,52,000, which is below 50,00,000, so round down → 7,00,00,000. Nearest ten-lakh (from the original 7,49,52,000): the lakhs digit is 9 and the part below ten-lakh is 9,52,000 ≥ 5,00,000, so round the ten-lakhs group up from 40 to 50 → 7,50,00,000. Rounding 49 lakh UP at the crore stage (wrong, since 49<50) gives 8,00,00,000. Rounding the ten-lakh DOWN wrongly gives 7,40,00,000.

28. (A) Round each: 4,82,000 → 5,00,000; 2,49,000 → 2,00,000 (49,000 < 50,000); 3,71,000 → 4,00,000. Sum of rounded values: 5,00,000 + 2,00,000 + 4,00,000 = 11,00,000. Rounding 4,82,000 down to 4,00,000 gives 10,00,000. Rounding 2,49,000 UP to 3,00,000 (wrong) gives 12,00,000. The exact sum 10,02,000 is not the round-first estimate the question asks for.

29. (C) 4,00,000 = 4×10^5; 25,00,000 = 25×10^5. Product = $4 \times 25 \times 10^{10} = 100 \times 10^{10} = 10^{12}$ = 1 followed by 12 zeros = 1,00,00,00,00,00,000, which has 13 digits. Forgetting that 100×10^{10} adds two more powers (carrying to 10^{12}) and stopping at 10^{11} gives 12 digits. Undercounting the zeros gives 11. Adding a spurious zero gives 14.

30. (C) Rounded factors: 50 passengers × 1,000 buses × 8 trips. Step by step: 50 × 1,000 = 50,000; 50,000 × 8 = 4,00,000. So about 4,00,000 passenger-trips a day. Dropping a zero gives 40,000. Adding a zero gives 40,00,000. Forgetting the 8 trips and using only 50 × 1,000 then a partial slip gives 80,000.

31. (B) Start 2,00,00,000. After draining: 2,00,00,000 − 78,40,000 = 1,21,60,000. After pumping back: 1,21,60,000 + 35,60,000 = 1,57,20,000. Stopping after the drain (forgetting the refill) gives 1,21,60,000. Subtracting both amounts gives 2,00,00,000 − 78,40,000 − 35,60,000 = 86,00,000-type slips; subtracting wrongly gives 1,14,00,000. Adding both to the start gives 2,42,00,000.

32. (D) Crores 8 → 8,00,00,000. 'eight thousand' = 8,000. 'eight' = 8. There is no lakh or ten-lakh part, so those slots are 0: 8,00,08,008. Dropping the empty crore-side structure collapses it to 8,08,008. Reading 'eight thousand' into the ten-thousand slot gives 8,00,80,008. Reading the lead 8 as ten-crore minus a period gives 80,08,008-type slips.

33. (D) Strip the commas: 7030400, which is 7 digits. Group Indian-style from the right (3, then 2, then 2): 400 | 30 | 70 → 70,30,400. The grouping 7,03,0400 uses a 4-digit lakh group and is wrong. The 2-2-2-2 split 7,03,04,00 mis-commas a 7-digit number. Keeping International-style commas on 7 digits gives 70,304,00, which is neither system.

34. (D) The leftmost digit 9 is in the crores place: place value 9,00,00,000. Reading the places: crores 9, ten-lakhs 0, lakhs 7, ten-thousands 4, thousands 0, hundreds 6, tens 0, ones 8. So the 4 is in the ten-thousands place: place value 40,000. Difference =

9,00,00,000 – 40,000 = 8,99,60,000. Reading the 4 as ten-lakhs (40,00,000) and subtracting gives 8,60,00,000. Reading it as lakhs (4,00,000) gives 8,96,00,000. Subtracting face values 9 – 4 gives 5.

35. (C) Group from the right: first 3 → 012, then 2 → 89, then 2 → 67, then 2 → 45, then 2 → 23, then 2 → 1. So 1,23,45,67,89,012. Applying a 2-then-rest split gives 12,34,56,78,9012. Pure 3-3-3 (International) gives 123,456,789,012. A 4-4-3 split gives 1,2345,6789,012.

36. (A) A 5-digit palindrome has the form a b c b a. Here a = 7 (so the ones digit is also 7) and the middle c = 3. The only free digit is b (the thousands and tens digit, which must match). b can be any of 0,1,2,...,9 — that is 10 choices. So there are 10 such palindromes (70307, 71317, ..., 79397). Forgetting that b = 0 is allowed gives 9. Treating b and c as independently free gives 10 × 10 = 100. Thinking the whole number is fixed gives 1.

37. (A) Convert all to plain numbers: 7.5 crore = 7,50,00,000; 76 million = 76 × 10,00,000 = 7,60,00,000; 7,55,00,000 = 7,55,00,000; 740 lakh = 740 × 1,00,000 = 7,40,00,000. Decreasing: 7,60,00,000 > 7,55,00,000 > 7,50,00,000 > 7,40,00,000, i.e. 76 million > 7,55,00,000 > 7.5 crore > 740 lakh. Treating 76 million as 7.55-crore-scale swaps the top two. Reading 740 lakh as 7.4 crore is correct but placing it on top reverses the list.

38. (B) Nearest ten-thousand: 6,95,000 has thousands digit 5, round up → 7,00,000. Nearest lakh (from original): ten-thousands digit is 9, round up → 7,00,000. Sum = 7,00,000 + 7,00,000 = 14,00,000. Rounding the ten-thousand down to 6,90,000 then adding 7,00,000 gives 13,90,000. Rounding the lakh down to 6,00,000 wrongly gives 13,00,000. Both down gives 12,90,000.

39. (A) Place the digits: 5 lakhs, 0 ten-thousands, 9 thousands, 0 hundreds, 7 tens, 0 ones → 5 × 1,00,000 + 0 × 10,000 + 9 × 1,000 + 0 × 100 + 7 × 10 + 0 × 1, which sums to 5,09,070. Putting the 9 in the ten-thousands place gives 5,90,700 (the form 5×1,00,000 + 9×10,000 + 7×100). Reading 5 as ten-lakhs gives 50,09,070. Putting the 7 in the hundreds place gives 5,09,700.

40. (C) Largest 3-digit number = 999; smallest 5-digit number = 10,000. Product = 999 × 10,000 = 99,90,000, which has digits 9-9-9-0-0-0-0 = 7 digits. Assuming a 3-digit times a 5-digit always gives 3 + 5 = 8 digits is the common slip giving 8. Over-counting gives 9. Counting only the non-zero front gives 5.

41. (C) 8,00,000 = 8 × 10^5. Product = 8 × 8 × 10^10 = 64 × 10^10 = 6,40,00,00,00,00,000? Count: 64 followed by 10 zeros = 64 then ten zeros = 12 digits total (6,4 and ten zeros = 2 + 10 = 12). So 12 digits. Treating 8 × 8 = 64 as a single digit and counting 11 is the slip. Adding an extra power gives 13. Counting only the zeros in one factor gives 10.

42. (D) 50,000 × 19,999 = 50,000 × 20,000 – 50,000 = 1,00,00,00,000 – 50,000 = 99,99,50,000. Count the digits: 9-9-9-9-5-0-0-0-0 = 9 digits. The careless estimate 50,000 × 20,000 = 10^9 would be 10 digits, but the true product is just under that, so it has only 9 digits — that overshoot is the trap giving 10. Under-counting gives 8; over-counting gives 11.

43. (A) Smallest 6-digit number = 1,00,000; smallest 4-digit number = 1,000. Product = 1,00,000 × 1,000 = 10,00,00,000, which is 1 followed by 8 zeros = 9 digits. Counting only the zeros (8) misses the leading 1, giving 8. Adding a spurious zero gives 10. Multiplying the digit counts 6 × 4 gives the nonsense 24.

44. (D) 8 lakh × 12 = 96 lakh = 96,00,000. 1 crore = 1,00,00,000. Since 96,00,000 < 1,00,00,000, one crore is larger (by 4 lakh). They are not equal. The claim '1 crore 6 lakh' miscounts 8 × 12 as 106. '1 crore is ten times larger' is false — it is only about 1.04 times larger.

45. (B) 0.65 crore = 65,00,000. 72 lakh = 72,00,000. Difference = 72,00,000 – 65,00,000 = 7,00,000 = 7 lakh. Treating the difference as 7 thousand drops two periods. Mistaking 0.65 crore as a near-2-lakh value gives a ~70 lakh gap. Reading 72 lakh as 12 lakh-over reverses the size and gives a 60 lakh-type error.

46. (B) Final = 1 crore 12 lakh = 1,12,00,000. Increase = 1,12,00,000 – 6,40,000 = 1,05,60,000. To the nearest ten-lakh: 1,05,60,000 lies between 1,00,00,000 and 1,10,00,000; the midpoint is 1,05,00,000, and 1,05,60,000 is above it, so round up to 1,10,00,000. Rounding down (treating the lakhs part as below half) gives 1,00,00,000. Forgetting to subtract the starting 6,40,000 fully gives a smaller increase near 90,00,000. Rounding only to the nearest lakh gives 1,06,00,000.

47. (D) Tripled total = 7,60,000 × 3 = 22,80,000. To the nearest ten-lakh: 22,80,000 lies between 20,00,000 and 30,00,000; the midpoint is 25,00,000; since 22,80,000 < 25,00,000, round down to 20,00,000. The exact total 22,80,000 is not a round banner figure. Rounding only to the nearest lakh gives 23,00,000. Rounding up to the next ten-lakh wrongly gives 30,00,000.

48. (B) Compare: 7,07,000 (seven lakh seven thousand) is the largest; 70,007 (seventy thousand seven) is the smallest; 7,00,070 is in the middle. Largest – smallest = 7,07,000 – 70,007 = 6,36,993. Subtracting the middle value 7,00,070 – 70,007 gives 6,30,063. Subtracting 7,07,000 – 7,070 (a comma slip) gives 6,99,930. Adding instead of borrowing correctly gives 7,36,993.

49. (D) Round 2,45,000 to the nearest lakh: 45,000 < 50,000, so round down → 2,00,000 litres/hour. Over 24 hours: 2,00,000 × 24 = 48,00,000 litres. Dropping a zero gives 4,80,000. Adding a zero gives 4,80,00,000. Rounding the rate UP to 3,00,000 (wrong, since 45<50) and multiplying gives ~72 lakh, and a partial slip toward that gives 60,00,000.

50. (C) The largest 10-digit number is 99,99,99,99,99 (ten 9s) and the smallest 11-digit number is 1,00,00,00,00,00 (1 followed by ten 0s). Consecutive integers differ by exactly 1 — the smallest 11-digit number comes immediately after the largest 10-digit number. Thinking the gap equals the next power difference gives 10 or a crore-scale figure. Believing they are the same gives 0.

51. (D) 1 million = 10,00,000 = 10 lakh. So 60 million = 60 × 10 lakh = 600 lakh. Treating a million as one lakh gives 60. Treating a million as a crore (100 lakh) gives 6,000. Treating 60 million as 60 lakh gives 6 (after a further slip).

52. (B) All start with 5 lakh; compare the next digits. 5,00,600 = five lakh six hundred (smallest). 5,06,000 = five lakh six thousand (middle). 5,60,000 = five lakh sixty thousand (largest). So increasing: 5,00,600 < 5,06,000 < 5,60,000. Misreading 5,60,000 as smaller than 5,06,000 swaps the top two. Treating 5,00,600 as the largest reverses the list.

53. (D) 54,000 × 10 = 5,40,000, so the wrong figure is 10 times too small (one period dropped). The absolute error is 5,40,000 − 54,000 = 4,86,000. Saying 100 times too small overcounts the dropped zeros. Reading the error as 4,86,00,000 misplaces two periods. Quoting the error as the full 5,40,000 forgets to subtract the 54,000 actually written.

54. (A) Mumbai = 1,28,00,000; Bengaluru = 84,00,000. Difference = 1,28,00,000 − 84,00,000 = 44,00,000. To the nearest ten-lakh: 44,00,000 lies between 40,00,000 and 50,00,000; midpoint 45,00,000; since 44,00,000 < 45,00,000, round down to 40,00,000. The exact difference 44,00,000 is not rounded to ten-lakh. Dropping a period gives 4,00,000. Rounding up wrongly gives 50,00,000.

55. (B) Weekly instalment = 1,00,00,000 ÷ 50 = 2,00,000. After 18 weeks, paid = 18 × 2,00,000 = 36,00,000. Still owed = 1,00,00,000 − 36,00,000 = 64,00,000. Reporting the amount PAID (36,00,000) instead of owed is the trap. Using 18 weeks remaining of 41 (a miscount) gives 82,00,000-type slips. Dropping a period gives 6,40,000.

56. (D) Round 4,950 → 5,000 and 6,050 → 6,000. Estimate = 5,000 × 6,000 = 5 × 6 × 10^6 = 30 × 10^6 = 3,00,00,000. The exact product 4,950 × 6,050 = 2,99,47,500, which is less than 3,00,00,000, so the estimate is more than the exact product. Thinking it is less ignores that 4,950 was rounded up by 50 while 6,050 was rounded down by only 50, and the net effect raises the estimate. Dropping a zero in the estimate gives 30,00,000. The estimate cannot equal the exact, since both factors were rounded.

57. (A) Total = 65,000 × 16 = 10,40,000. To the nearest lakh: the part below the lakh is 40,000 < 50,000, so round down → 10,00,000. The exact total 10,40,000 is not yet rounded. Rounding up wrongly gives 11,00,000. A digit-count slip gives 1,00,000.

58. (B) 3,49,500: the part below the lakh is 49,500 < 50,000, so round down → 3,00,000. 3,50,001: the part below the lakh is 50,001 ≥ 50,000, so round up → 4,00,000. Treating both as below half gives 3,00,000 twice. Treating both as at/above half gives 4,00,000 twice. Swapping the directions gives 4,00,000 and 3,00,000.

59. (C) Digits: crores 6, ten-lakhs 0, lakhs 3, ten-thousands 5, thousands 0, hundreds 0, tens 7, ones 0. The 3 is in the lakhs place: place value 3,00,000. The face value of the 5 is just 5. So 3,00,000 ÷ 5 = 60,000. Treating the divisor as the 5's place value (50,000) gives 3,00,000 ÷ 50,000 = 6 then a comma slip lands on 6,00,000. Forgetting to divide at all leaves a 3,00,000-scale slip of 60,00,000. Dividing 3,00,000 by 25 gives 12,000.

60. (C) Number of doctors ≈ total people ÷ people-per-doctor = 5,00,000 ÷ 1,250. Compute 5,00,000 ÷ 1,250 = 5,00,000 ÷ 1,250 = 400 (since 1,250 × 400 = 5,00,000). So about 400 doctors. Dropping a zero in the division gives 4,000. Dividing by 12,500 instead of 1,250 gives 40. Multiplying 5,00,000 by a wrong ratio, or using 800 people-per-doctor, gives about 625.